

1. Data Overview

The “Supermarket Sale” dataset contains retail transaction data commonly used for business analytics and educational analysis purposes. The dataset represents customer purchasing activities across supermarket branches in different cities and includes information related to customer groups, product categories, purchase quantity, and transaction value.

I chose this dataset because it provides a practical business context and allows meaningful analysis of customer behaviour and sales performance. Since the dataset includes both categorical and numerical variables, it is suitable for Exploratory Data Analysis (EDA) using Excel tools such as PivotTables, descriptive statistics, sorting, and data visualisation.

Size: 253 rows × 8 columns

- **Sale_ID**: Unique identifier for each sales transaction.
- **Branch**: The supermarket branch where the transaction occurred.
- **City**: The city where the customer made the purchase.
- **Customer_Type**: Customer classification, including Member, Normal, and Unknown customers.
- **Product_Name**: Name of the purchased product.
- **Product_Category**: Product category such as Fruits, Beverages, Household, Personal Care, and Stationery.
- **Quantity**: Number of product units purchased in each transaction.
- **Total_price**: Total transaction value in USD.

Using these variables, I focused mainly on analysing customer spending behaviour and identifying which product categories contribute the most to supermarket revenue. These findings are useful because they can support business decisions related to customer retention, inventory planning, and promotional strategies.

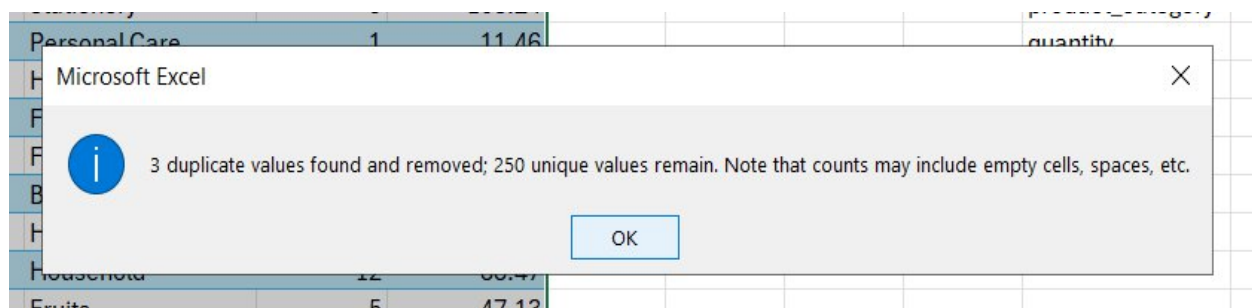
	B	C	D	E	F	G	H
1	branch	city	customer_type	product_name	product_category	quantity	total_price
2	A	New York	Member	Shampoo	Personal Care	3	17.66
3	B	Los Angeles	Normal	Notebook	Stationery	10	29.43
4	A	New York	Member	Apple	Fruits	15	19.26
5	A	Chicago	Normal	Detergent	Household	5	41.73
6	B	Los Angeles	Member	Orange Juice	Beverages	7	26.22
7	A	Chicago	Normal	Shampoo	Stationery	9	108.24
8	A	Chicago	Normal	Shampoo	Personal Care	1	11.46
9	B	Los Angeles	Normal	Shampoo	Household	9	175.55
10	A	Chicago	Member	Apple	Fruits	20	302.81
11	B	Los Angeles	Member	Shampoo	Fruits	19	374.48
12	A	Chicago	Normal	Detergent	Beverages	7	69.81
13	A	New York	Normal	Orange Juice	Household	4	14.08
14	B	Los Angeles	Member	Orange Juice	Household	12	88.47
15	A	New York	Normal	Orange Juice	Household	4	14.08
16	B	Los Angeles	Member	Apple	Fruits	5	47.13
17	B	Los Angeles	Normal	Apple	Beverages	3	62.53
18	B	Los Angeles	Normal	Notebook	Stationery	8	47.51
19	B	Los Angeles	Member	Apple	Personal Care	2	4.56
20	B	Los Angeles	Normal	Notebook	Fruits	15	212.82
21	A	New York	Member	Apple	Beverages	2	41.92
22	A	Chicago	Normal	Detergent	Personal Care	11	59.75
23	B	Los Angeles	Normal	Shampoo	Personal Care	11	51.32
24	B	Los Angeles	Normal	Orange Juice	Beverages	1	7.28
25	A	New York	Member	Shampoo	Beverages	16	33.38
26	A	Chicago	Normal	Apple	Fruits	2	40.77
27	B	Los Angeles	Member	Detergent	Beverages	18	218.22

2. Data Cleaning

Before starting the analysis, I reviewed the dataset carefully to identify any issues that could affect the accuracy and reliability of the results. In business analytics, data cleaning is an important step because duplicate or incomplete data may distort statistical results and lead to misleading business insights.

Handling Duplicate Records

I first checked the dataset for duplicate transactions by using Excel's "Remove Duplicates" function. During this process, I identified 3 duplicate rows in the dataset. Since duplicated transactions could artificially increase total revenue and affect descriptive statistics, I removed all duplicated records before conducting further analysis.



Data Cleaning Result	Value
Original number of rows	253
Duplicate rows removed	3
Final cleaned rows	250

After removing duplicates, the final cleaned dataset had 250 records. Although the number of duplicates was relatively small, removing them was still necessary to improve the reliability of the analysis.

Handling Missing Values

After checking the dataset carefully in Excel, I found blank cells in three variables: customer_type, product_category, and quantity.

Column	Number of Blank Cells	Handling Method
customer_type	3	Replaced with “Unknown”
product_category	6	Replaced with “Unknown”
quantity	3	Replaced using median value

Customer Type

The customer_type column contained 3 blank cells. Instead of deleting these records, I replaced the missing values with “Unknown”. I decided to keep these transactions because the remaining information, such as purchased products, quantity, and total transaction value, was still complete and useful for the analysis.

From a business perspective, incomplete customer information may occur when customers do not register membership details or when transaction information is entered incorrectly during checkout. Therefore, keeping these records makes the dataset more realistic while preserving important sales information.

	A	B	C	D	E	F	G	H
1	sale_id ▼	branch ▼	city ▼	customer_type ▼	product_name ▼	product_category ▼	quantity ▼	total_price ▼
34	31	A	Chicago		Shampoo	Fruits	3	11,88
38	35	A	New York		Apple	Personal Care	2	28,72
48	44	A	Chicago		Detergent	Fruits		43,08
255								
256								

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
34	31	A	Chicago	Unknown	Shampoo	Fruits	3	11,88
38	35	A	New York	Unknown	Apple	Personal Care	2	28,72
48	44	A	Chicago	Unknown	Detergent	Fruits		43,08
255								

Product Category

The product_category column contained 6 blank cells. Similar to customer_type, I replaced these missing values with “Unknown” rather than assigning them randomly to another category.

I selected this approach because there was no reliable information showing which category the products originally belonged to. Randomly assigning categories could create inaccurate results and reduce the reliability of the product category analysis.

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
33	30	A	Chicago	Member	Orange Juice		15	198,54
35	32	A	New York	Member	Orange Juice		3	17,3
53	49	B	Los Angeles	Normal	Detergent		2	13,5
72	68	A	New York	Normal	Detergent		1	7,22
91	87	A	Chicago	Member	Shampoo		2	5,86
103	99	A	New York	Normal	Detergent		17	277,4
255								

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
33	30	A	Chicago	Member	Orange Juice	Unknown	15	198,54
35	32	A	New York	Member	Orange Juice	Unknown	3	17,3
53	49	B	Los Angeles	Normal	Detergent	Unknown	2	13,5
72	68	A	New York	Normal	Detergent	Unknown	1	7,22
91	87	A	Chicago	Member	Shampoo	Unknown	2	5,86
103	99	A	New York	Normal	Detergent	Unknown	17	277,4
255								

Quantity

The quantity column contained 3 blank cells. Since quantity is a numerical variable, replacing blank values with text labels would not be appropriate. To handle this issue, I calculated the median quantity in Excel using the following formula: =MEDIAN(G:G). The median value was 11, so I replaced all blank quantity cells with 11.

1	branch	city	customer_type	product_name	product_category	quantity	total_price
22	A	Chicago	Normal	Detergent	Personal Care		59.75
48	A	Chicago	Unknown	Detergent	Fruits		43.08
65	A	New York	Member	Detergent	Household		20.09
255							

	B	C	D	E	F	G	H	I
1	branch	city	customer_type	product_name	product_category	quantity	total_price	
22	A	Chicago	Normal	Detergent	Personal Care	11	59.75	
48	A	Chicago	Unknown	Detergent	Fruits	11	43.08	
65	A	New York	Member	Detergent	Household	11	20.09	
255								

I chose the median instead of the mean because the median is less affected by unusually high or low values and better represents the typical purchasing quantity within the dataset. Using the median helps maintain the stability of the data and reduces the influence of outliers on the analysis.

After handling duplicate records and missing values, the dataset became cleaner and more suitable for descriptive statistics, PivotTable analysis, and visualisation.

3. Descriptive Statistics

After completing the data cleaning process, I used descriptive statistics to summarise the main characteristics of the dataset and better understand customer purchasing patterns. The analysis focused mainly on two numerical variables: quantity and total_price.

Quantity Statistics

quantity	
Mean	10.61264822
Standard Error	0.375926392
Median	11
Mode	2
Standard Deviation	5.97947532
Sample Variance	35.7541251
Kurtosis	-1.257327406
Skewness	-0.072949934
Range	19
Minimum	1
Maximum	20
Sum	2685
Count	253

The average quantity purchased per transaction is approximately 10.62 units, while the median quantity is 11 units. Since the mean and median are very close, the purchasing quantity distribution appears relatively balanced.

However, the standard deviation of 5.98 indicates that customer purchasing quantity still varies noticeably across transactions. Some customers purchase only a few items during a visit, while others buy much larger quantities depending on their shopping needs.

This result suggests that customer purchasing behaviour is not consistent across all transactions, which is common in supermarket environments where customers have different consumption habits and spending capacities.

Total Price Statistics

	<i>total_price</i>
Mean	123.7659289
Standard Error	6.481325292
Median	94.17
Mode	14.08
Standard Deviation	103.0917898
Sample Variance	10627.91712
Kurtosis	0.033590636
Skewness	0.907731819
Range	424.96
Minimum	2.18
Maximum	427.14
Sum	31312.78
Count	253

The total_price variable shows a larger variation in customer spending behaviour. The average transaction value is approximately 123.77, while the median transaction value is lower at 94.17.

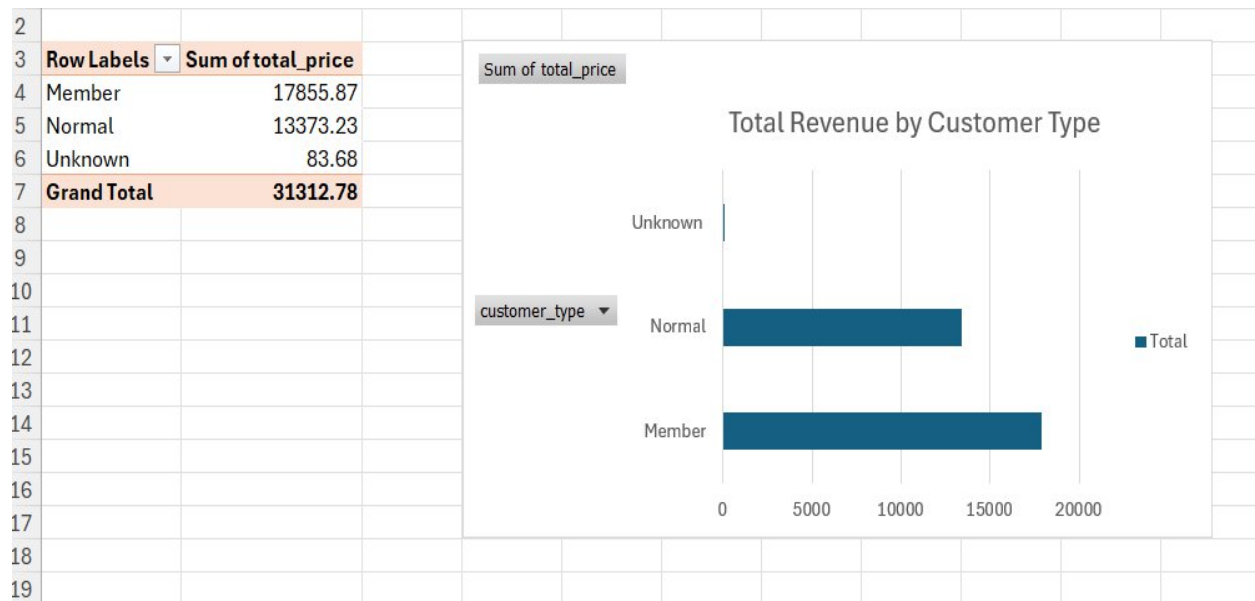
The noticeable gap between the mean and median suggests that several high-value purchases increased the overall average transaction amount. This indicates that although many customers spend around the median level, a smaller number of customers contribute significantly larger transaction values. In addition, the standard deviation is relatively high compared with the mean, showing substantial differences in spending behaviour between customers. Some customers make small routine purchases, while others spend much more during a single shopping trip.

From a business perspective, this finding highlights that revenue contribution is not evenly distributed across all customers. Certain customer groups and purchasing behaviours contribute more strongly to total sales performance.

4. Insights

Insight 1: Member Customers Generate Stronger Revenue Performance

To better understand customer spending behaviour, I used a PivotTable and bar chart to compare total revenue generated by different customer groups.



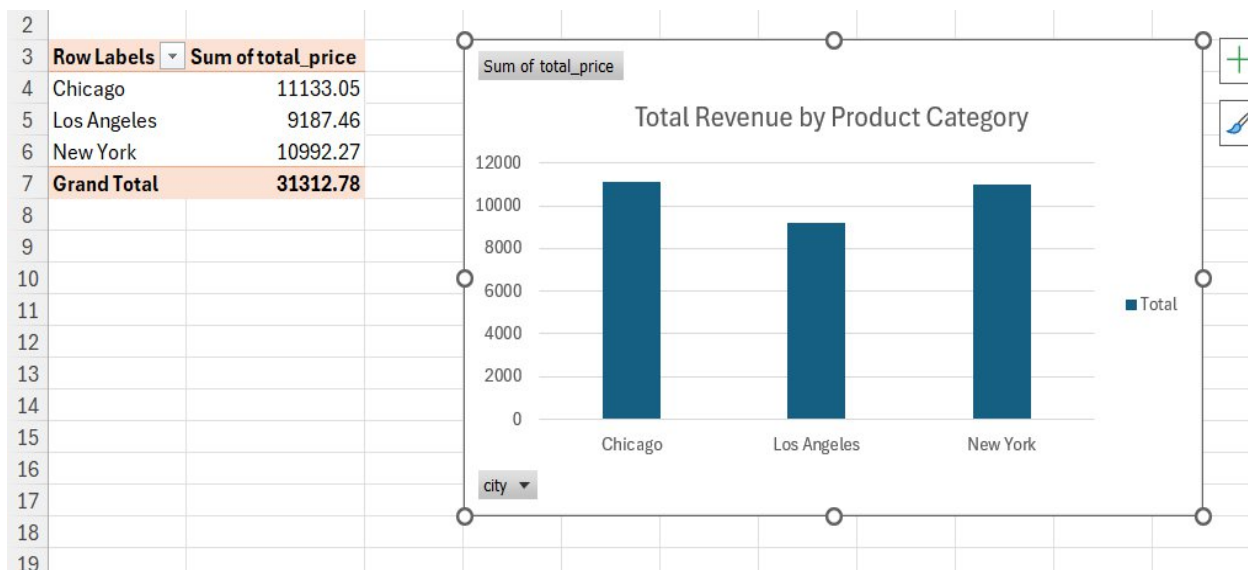
The analysis shows that Member customers generated significantly higher revenue compared with Normal customers. Member customers contributed approximately 57% of the total revenue, making them the most valuable customer segment in the dataset.

Besides total revenue, I also compared the average spending value per transaction. Member customers spent approximately 136.30 per transaction, while Normal customers spent around 112.99. This indicates that Member customers not only shop more regularly, but also tend to spend more during each purchase. From my perspective, this result strongly suggests that loyalty programs positively influence customer purchasing behaviour. Customers who receive benefits such as discounts, reward points, or exclusive promotions are more likely to remain loyal and continue purchasing from the supermarket.

This insight is important because retaining existing customers is often more cost-effective than continuously attracting new customers. Based on these findings, the supermarket should continue improving its membership program by offering stronger reward systems, personalised promotions, and exclusive member benefits to increase long-term customer value. Moreover, the small revenue contribution from the “Unknown” customer group also highlights a minor data recording issue. Although the impact is relatively small in this dataset, incomplete customer information may reduce the accuracy of future customer segmentation analysis.

Insight 2: Fruits Category Generates the Highest Sales Performance

To evaluate product category performance, I analysed total revenue generated by each category using a PivotTable and category sales chart.



The analysis shows that the Fruits category generated the highest total revenue among all product categories. Fruits contributed approximately 25% of total category revenue, indicating strong and stable customer demand for food-related products.

One possible explanation is that fruits are daily necessity products purchased more frequently compared with categories such as Personal Care or Household products. Since these products are consumed regularly, customers tend to purchase them repeatedly during routine shopping trips. Also, this data shows that Beverages and Stationery generated relatively strong sales performance. This suggests that customers often purchase products from multiple categories during the same visit rather than purchasing only one type of product.

From a business perspective, this insight can support inventory management and sales strategy decisions. High-performing categories such as Fruits and Beverages should receive greater inventory attention and stronger promotional support to avoid stock shortages and maintain sales performance. In addition, the supermarket could apply cross-selling strategies by placing related products near each other to encourage customers to increase their spending during each shopping trip.

Overall, this analysis demonstrates how product category analysis can help businesses better understand customer purchasing priorities and improve operational decision-making related to inventory planning, product placement, and revenue optimisation.